MCO-1D Electronics Bill of Materials

Metachronal Cascade Ornithopter — Drone Prototype · Locked specification Rev H · Issue date: 2026-06-03

This BOM covers electronics only. Not included: airframe structure (carbon tube, printed parts, bearings), wing materials (LW-PLA filament, CF rods, CF skin, ripstop nylon membrane), tools, LiPo charger, or adhesives. Prices are approximate GBP at mid-2026 market rates. AliExpress sourcing is 30-40% cheaper but allow 3-4 weeks delivery.

Flap Drive

The drive system that makes the wings flap. Eight brushless motors (one per wing) driven by eight ODrive Micro motor controllers talking over CAN-FD. The 7:1 rope reducer converts motor rotation to wing flap at the correct frequency.

Item	Specification	Qty	Unit £	Total £	Notes	Source
GoolRC 5010 360KV Brushless Motor	5010 frame · 360KV · 90g · bullet connectors	8	£12.50	£100.00	One per wing. Confirmed locked Rev H specification.	AliExpress / Amazon search: GoolRC 5010 360KV
ODrive Micro motor controller	ODrive Micro · 6S compatible · CAN-FD bus	8	£28.00	£224.00	One per motor. CAN node IDs 0-7. Controls velocity closed-loop.	odriverobotics.com ODrive Micro
Dyneema rope 1.8mm	UHMWPE braid · 1.8mm dia · ~4000N break strength	2	£8.00	£16.00	~1m per wing = ~8m total. Buy two 5m reels.	Fishing / kite shop Dyneema SK75 or SK99
Drive pulley 12mm OD	12mm OD · 3mm bore · groove for 1.8mm rope	8	£2.50	£20.00	Motor shaft pulley. PETG print acceptable for prototype.	Custom print or machine One per motor
Driven pulley 84mm OD	84mm OD · groove for 1.8mm rope · 7:1 with 12mm drive	8	£4.00	£32.00	Wing crank pulley. 84÷12 = 7:1 reducer ratio.	Custom print or machine One per wing

Subtotal: £392.00

Servos

Two servos per wing: one for sweep (how far the wing is swept back) and one for pitch (the wing's angle of attack). 16 servos total.

Item	Specification	Qty	Unit £	Total £	Notes	Source
ZOSKAY DS3225 25KG Digital Servo	DS3225 · 25KG torque · 270° range · 60g · 2.40 N·m	8	£9.50	£76.00	Sweep axis. 270° range essential — sweeps wing 0°→210°. Confirmed Rev H.	Amazon / AliExpress search: DS3225 25KG servo
Emax ES08MD II Digital Micro Servo	ES08MD II · digital · 2.1KG · 8.5g · 0.08s/60°	8	£6.00	£48.00	Pitch axis. Controls wing angle of attack. Confirmed Rev H.	Hobbyking / Amazon Emax ES08MD II

Subtotal: £124.00

Computers

Two computers working together. The Raspberry Pi runs the wave library — all the flapping patterns. The Pixhawk is the safety co-pilot that reads all the sensors and will take over if anything goes wrong.

Item	Specification	Qty	Unit £	Total £	Notes	Source
Raspberry Pi Zero 2W	RPi Zero 2W · 512MB RAM · WiFi · 40-pin GPIO · USB-C power	1	£15.00	£15.00	Wave library CPU. Runs all firmware. Talks to ODrives via CAN-FD.	raspberrypi.com or approved resellers
Pixhawk Mini flight controller	Pixhawk Mini · STM32F4 · built-in IMU · barometer · JST-GH ports	1	£55.00	£55.00	Safety FCC. Reads all sensors. Sends MAVLink to RPi. Runs ArduPlane.	Holybro.com or GetFPV Pixhawk Mini

Subtotal: £70.00

CAN-FD Interface

The Raspberry Pi Zero 2W does not have CAN-FD built in. This small add-on chip gives it the ability to talk to all 8 ODrive motor controllers over a single 4-wire bus.

Item	Specification	Qty	Unit £	Total £	Notes	Source
MCP2518FD SPI-to-CAN-FD module	MCP2518FD breakout · SPI interface · 5V tolerant · CAN-FD 5Mbit/s	1	£8.00	£8.00	Connects to RPi via SPI (4 wires). Essential for ODrive communication.	AliExpress search: MCP2518FD breakout
CAN bus termination resistors 120Ω	120Ω · 0.25W · through-hole	2	£0.10	£0.20	One resistor at each end of CAN bus. Without these the bus is unreliable.	RS / Mouser / eBay any electronics supplier
CAN bus cable — 4-wire	22AWG or 24AWG twisted pair · ~2m needed	1	£3.00	£3.00	Runs between MCP2518FD and all 8 ODrives. Can repurpose CAT5 ethernet cable.	Electronics supplier or CAT5 offcuts

Subtotal: £11.20

Power System

Everything runs off one 6-cell lithium polymer battery. The power flows through a distribution board to the motors, and through a voltage regulator to the computers and servos. A current sensor tells the Pixhawk how fast the battery is draining.

Item	Specification	Qty	Unit £	Total £	Notes	Source
6S LiPo battery 5000mAh	6S · 22.2V nominal · 5000mAh · 50C+ discharge · XT60 connector	1	£55.00	£55.00	Confirmed locked Rev H. ~20 min cruise endurance. Buy a LiPo-safe bag too.	Hobbyking / RDQ / Tattu 6S 5000mAh 50C LiPo
XT60 connector pairs	XT60 · gold plated · male + female pairs	4	£0.80	£3.20	Main battery connector. Buy spares — you will solder several.	Any FPV / RC supplier
5V 3A BEC (voltage regulator)	Input: 5S-8S · Output: 5V 3A regulated · switching type	1	£6.00	£6.00	Converts 22V battery down to 5V for the RPi and all 16 servos.	Pololu D24V22F5 or Matek UBEC 5V 3A
Pixhawk power module with current sensor	0-90A · 6S compatible · XT60 in/out · JST-GH to Pixhawk	1	£12.00	£12.00	Measures battery current and voltage. Sends data to Pixhawk for monitoring.	Holybro PM02D or Matek FCHUB
Power distribution board	6S input · 8× outputs · XT60 or solder pads	1	£8.00	£8.00	Splits battery supply to all 8 ODrives. Matek FCHUB or copper bus bars.	Matek FCHUB-6S or custom copper bars
Capacitors 470µF 50V low-ESR	Electrolytic · 470µF · 50V · low ESR	2	£1.50	£3.00	Mounted close to ODrives. Smooths voltage spikes when motors start/stop.	RS / Mouser Panasonic FR series

Subtotal: £87.20

Sensors

Three sensors tell the aircraft what is happening in the world around it: how fast the air is moving (airspeed), how far from the ground it is (rangefinder), and where on Earth it is (GPS).

Item	Specification	Qty	Unit £	Total £	Notes	Source
Pitot tube + airspeed sensor	MS4525DO · I2C · 5V · includes pitot tube and silicone tubing	1	£15.00	£15.00	Mounts on nose. Measures airspeed for stall protection and speed control. Essential.	Matek ASPD-4525 or RCSchim pitot set
TF-Luna lidar rangefinder	TF-Luna · 0.2–8m · ±2cm · 250Hz · UART or I2C · 5g	1	£18.00	£18.00	Points straight down from undercarriage. Controls wing amplitude near ground.	Benewake TF-Luna Amazon / AliExpress
GPS module with compass	M8N or M9N · GPS+GLONASS · built-in compass · Pixhawk connector	1	£35.00	£35.00	Mount on short mast away from motors. Return-to-home if radio link lost.	Holybro M8N GPS holybro.com

Subtotal: £68.00

RC System

The radio link between your hand controller and the aircraft. The receiver on the aircraft picks up your stick movements and sends them to the Pixhawk.

Item	Specification	Qty	Unit £	Total £	Notes	Source
RC receiver — SBUS output	FrSky XM+ · SBUS · 8 channels · 2.4GHz	1	£18.00	£18.00	Must match your transmitter brand/protocol. SBUS connects to Pixhawk RC port.	FrSky XM+ for FrSky TX or BETAFPV ELRS for ExpressLRS TX
RC transmitter (hand controller)	8+ channels · telemetry · FrSky or ELRS protocol	1	£140.00	£140.00	Your hand controller. Not a drone component but listed for budget. Radiomaster TX16S is good value with both FrSky and ELRS support.	Radiomaster TX16S or FrSky Taranis X9D

Subtotal: £158.00

Telemetry

A wireless link between the aircraft and your laptop or tablet on the ground. Lets you watch sensor readings, airspeed, GPS position and battery level in real time while flying. Not essential but very useful for first flights.

Item	Specification	Qty	Unit £	Total £	Notes	Source
433MHz SiK telemetry radio pair	SiK radio · 433MHz · 100mW · USB ground station + aircraft module	1	£28.00	£28.00	Ground module plugs into laptop USB. Aircraft module to Pixhawk TELEM1.	Holybro SiK Telemetry V3 or RFDesign RFD900

Subtotal: £28.00

Wiring and Connectors

All the cables, connectors, and small boards needed to wire everything together. Easy to underestimate — buy more than you think you need.

Item	Specification	Qty	Unit £	Total £	Notes	Source
Servo extension cables 30cm	26AWG · JR/Futaba 3-pin · 30cm length	20	£0.60	£12.00	16 needed for servos + 4 spares. Can cut to length from bulk reel.	Hobbyking servo wire or any RC supplier
PCA9685 16-channel PWM board	PCA9685 · I2C · 16 channels · 5V · for servo PWM signals	1	£4.00	£4.00	Optional but recommended. More reliable than direct GPIO PWM for 16 channels.	AliExpress search: PCA9685 servo driver
Silicone wire 16AWG red + black	Silicone insulated · 16AWG · 1m of each colour	2	£4.50	£9.00	Main power wiring battery → distribution → ODrives. Must be silicone not PVC.	HobbyKing / RCSchim silicone wire 16AWG
Silicone wire 22AWG mixed	Silicone insulated · 22AWG · assorted colours · ~5m	1	£5.00	£5.00	Signal wiring: CAN bus, I2C, UART connections between boards.	Any electronics supplier
JST-GH connector kit	JST-GH 1.25mm · 4-pin and 6-pin · crimp housings + pins	1	£8.00	£8.00	All Pixhawk ports use JST-GH. Buy a full kit — you will make many cables.	Holybro connector kit or AliExpress JST-GH kit
Heat shrink tubing assortment	2mm · 4mm · 6mm · black · mixed lengths	1	£4.00	£4.00	Insulation and strain relief on all soldered joints.	Any electronics supplier

Subtotal: £42.00

Mounting Hardware

Small hardware items needed to mount all the electronics securely and safely inside the airframe.

Item	Specification	Qty	Unit £	Total £	Notes	Source
Anti-vibration mounts for Pixhawk	M3 silicone vibration isolators or 3M foam pads	1	£4.00	£4.00	Critical — Pixhawk IMU is ruined by vibration. Use foam or rubber mounts.	Included with Pixhawk Mini or buy 3M VHB foam separately
Nylon standoffs M2 and M3	M2×10mm and M3×6mm nylon · mixed pack	1	£3.50	£3.50	Board mounting for RPi, PCA9685, MCP2518FD. Nylon avoids shorts.	eBay: nylon standoff kit
MicroSD card 32GB	MicroSD · Class 10 / U1 · 32GB	1	£8.00	£8.00	RPi boot drive. Pre-load with Raspberry Pi OS Lite + firmware files.	SanDisk or Samsung any electronics retailer
USB-C cable 20cm	USB-C · 20cm · for BEC to RPi power connection	1	£3.00	£3.00	Connects 5V BEC output to RPi power input.	Any supplier

Subtotal: £18.50

Cost Summary

Category	Subtotal
Flap Drive	£392.00
Servos	£124.00
Computers	£70.00
CAN-FD Interface	£11.20
Power System	£87.20
Sensors	£68.00
RC System	£158.00
Telemetry	£28.00
Wiring and Connectors	£42.00
Mounting Hardware	£18.50
[redacted]	[redacted]
Total excluding RC transmitter	**£858.90**
RC transmitter (Radiomaster TX16S)	**£140.00**
GRAND TOTAL	**£998.90**

MCO-1D Electronics BOM · Issue 1.0 · 2026-06-03 · 34 line items · All items cross-checked against locked Rev H specification. Prices approximate GBP mid-2026. Not included: airframe, wing materials, tools, charger.